



Applied Chemistry

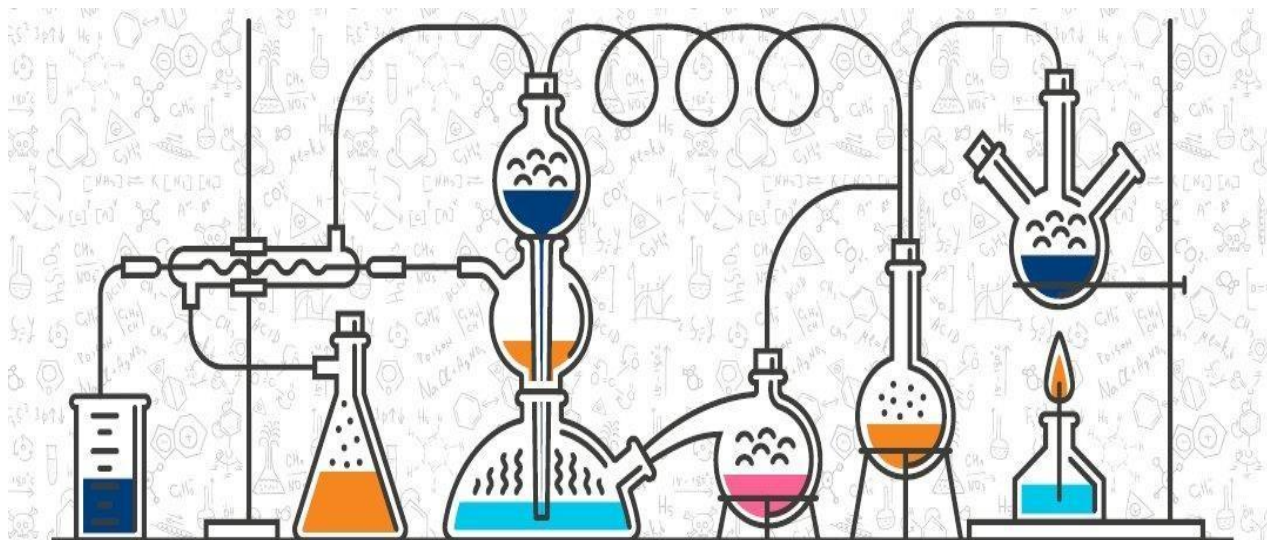
Chemistry of Smart Energy Systems- CS Stream

Stream: CS (Common to CS, IS, AI, CD, CY, CG, IC & BT Programs)

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Module - V



DAYANANDA SAGAR COLLEGE OF ENGINEERING

Accredited by National Assessment & Accreditation Council (NAAC) with 'A' Grade
(An Autonomous Institution affiliated to Visvesvaraya Technological University, Belagavi &
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DEPARTMENT OF CHEMISTRY

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- Solubility: Soluble in chlorinated solvents, dioxane, benzene, and THF.
- Biodegradable and biocompatible, suitable for human contact applications.

Uses of PLA in Touch Screen Applications

- **Flexible Substrate:** PLA films can be used as a lightweight, flexible substrate for touch screens.
- **Eco-friendly Material:** Reduces environmental impact compared to petroleum-based plastics.
- **Protective Coating:** Transparent PLA coatings improve scratch resistance.
- **Printable Surface:** PLA allows for easy deposition of conductive layers (like ITO or silver nanowires) for capacitive touch sensors.
- **Biocompatibility:** Safe for devices with frequent human contact.

Polyethylene Glycol (PEG)

Introduction:

PEG is a hydrophilic polymer made from ethylene oxide. Known for solubility in water and organic solvents, ethylene glycol, or other initiators.

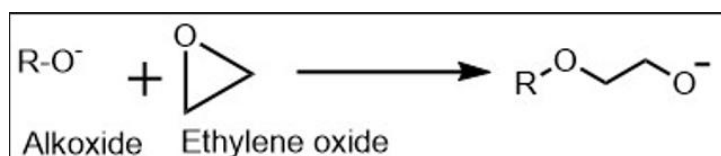
Synthesis of PEG

Step 1: Initiation

An alcohol (R-OH) is converted into an alkoxide initiator (R-O⁻) using a base.



The alkoxide attacks the ethylene oxide ring: The three-membered epoxide ring opens because of ring strain.

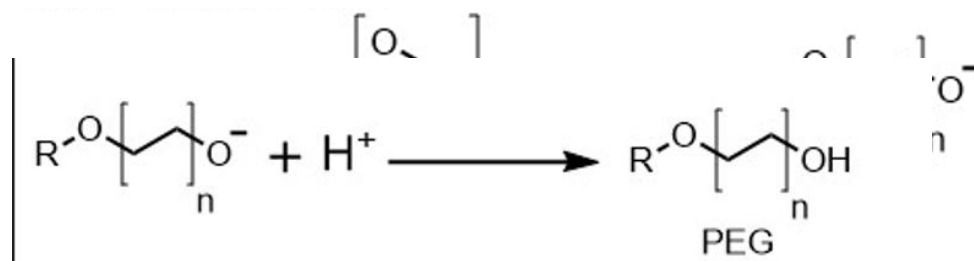


Step 2: Propagation

The newly formed alkoxide continues attacking additional ethylene oxide molecules and the process continues

Step 3: Termination (Acid Work-Up)

After the desired chain length is reached, dilute acid is added to give the final PEG polymer



Properties of PEG:

- Hydrophilic polymer, soluble in water and many organic solvents.
- Flexible chain; low glass transition temperature (~ -60°C).
- Melting point: Depends on molecular weight (solid for high MW, liquid for low MW).
- Non-toxic, biocompatible, and chemically stable.
- Low viscosity for low molecular weight PEG; high viscosity for high molecular weight PEG.

Uses of PEG in Touch Screen Applications:

- **Flexible Coatings:** PEG can be used as a transparent, flexible layer to protect touch screens.
- **Anti-fouling Coatings:** Prevents fingerprint smudges and dust accumulation.
- **Surface Modifier:** Enhances adhesion of conductive layers (ITO, silver nanowires) on flexible substrates.
- **Hydrophilic Layer:** Improves moisture management in flexible or foldable touch panels.

- **Electrolyte Component:** Used in solid polymer electrolytes for capacitive touch sensors and flexible electronics.

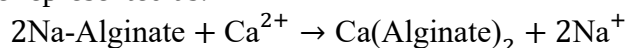
Synthesis and properties of Alginate Hydrogel and its uses in Brain Computer Interfaces (BCIs) applications.

Alginate hydrogel is a three-dimensional, water-swollen polymeric network formed from alginate, a natural polysaccharide extracted mainly from brown seaweed. Alginate consists of β -D-mannuronic acid (M) and α -L-guluronic acid (G) units. Due to its biocompatibility, biodegradability, non-toxicity, and easy gelation, alginate hydrogel is widely used in biomedical, environmental, and catalytic applications.

Synthesis of Alginate Hydrogel (Ionotropic Gelation)

- **Preparation of solution:** Sodium alginate, a linear polysaccharide from brown seaweed dissolved in water to form the solution.
- **Cross-linking Process:** A solution of divalent cations, typically CaCl_2 , is added. Calcium ions (Ca^{2+}) replace sodium ions (Na^+) in G-blocks of alginate chains.
- **Gel formation :**

The gelation reaction can be represented as:



The calcium ions interact mainly with guluronic acid blocks, forming the well-known "**egg-box**" structure.

- **Washing and Drying**

Wash the formed hydrogel with distilled water to remove excess CaCl_2 .

Use directly as a wet hydrogel or dry at room temperature/oven for xerogel formation.

Properties of Alginate Hydrogel:

- **Biocompatibility & Biodegradability:** Supports cell adhesion and growth.
- **High water content & Porosity:** Facilitates nutrient and ion transport.
- **Ionic Conductivity:** Hydrated ions allow electrical signal conduction.
- **Tissue-like Mechanical Properties:** Soft, flexible, tunable stiffness.
- **Low Immunogenicity:** Minimal inflammatory response; resistant to protein adsorption.
- **Electrochemical Adaptability:** Can form conductive composites with polymers/nanoparticles.

Applications in Brain-Computer Interfaces (BCIs):

- **Soft Electrode Interface:** Conforms closely to brain tissue, reducing inflammation and motion artifacts.
- **Signal Enhancement:** Ionic conductivity and integration with conductive materials improve signal-to-noise ratio (SNR) for electrophysiological recordings.
- **Chronic Implant Stability:** Supports long-term recording by reducing tissue scarring and immune response.
- **Flexible, Implantable Electrodes:** Used in electrocorticography (ECoG) and local field potential (LFP) recording.
- **Hybrid Hydrogel Systems:** Can incorporate conductive polymers/nanoparticles to enhance electrical properties while maintaining softness.
- Can be engineered for transparency, enabling optical and electrical monitoring.

Introduction for Sensors: A sensor is a device that receives a signal (i.e. physical, chemical or biological signal) and responds to the signal in the form of an electrical signal. The output signals correspond to some forms of electrical signal, such as current or voltage.

Components of a sensor

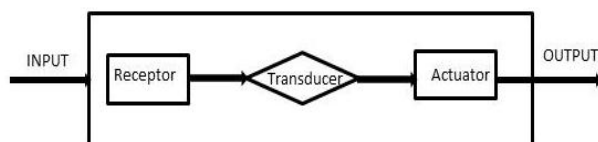


Figure 1: Function of a sensor

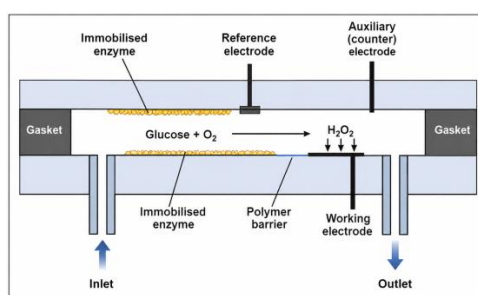
1. Receptor- Receives/ observes and acquires information.
2. Transducer-Converts one form of energy into another form.
3. An Actuator-Converts electrical energy into mechanical/ readable energy.

Principle and Working Mechanism for Detection of Glucose in Biofluids:

A **glucose biosensor** is an analytical device used to determine the concentration of glucose in biological fluids such as blood, saliva, sweat, tears, and urine. It consists of a **bioreceptor (enzyme)**, a **transducer**, and a **signal processing unit**.

Principle

The glucose biosensor works on the principle of a specific enzymatic reaction between glucose and the enzyme **Glucose Oxidase (GOx)**. The enzyme selectively oxidizes glucose, producing gluconic acid and hydrogen peroxide. The transducer usually a working electrode, converts the resulting biochemical reaction into an electrical signal, which is proportional to the glucose concentration.



Working Mechanism

1. The biofluid sample containing glucose is placed on the biosensor surface.
2. Glucose diffuses to the enzyme layer containing immobilized **Glucose Oxidase (GOx)**.
3. The enzyme catalyzes the oxidation of glucose according to the reaction:

$$\text{Glucose} + \text{O}_2 \rightarrow \text{Gluconic Acid} + \text{H}_2\text{O}_2$$
4. The hydrogen peroxide produced reaches the **working electrode (transducer)** where it undergoes electrochemical oxidation:

$$\text{H}_2\text{O}_2 \rightarrow \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$$
5. The released electrons are collected by the working electrode, generating an electrical current.
6. The magnitude of the current is directly proportional to the amount of glucose present in the biofluid.
7. A **reference electrode** maintains a constant potential, while the transducer converts the electrochemical signal into an electrical signal.
8. The signal processing unit amplifies and processes the signal, and the glucose concentration is displayed on the readout device.

CONDUCTOMETRIC TITRATIONS:

Conductometric sensor: Estimation of acid mixture

Principle: The electrical conductance depends on the concentration of solution and mobility of ions.

Analysis based on measurements of conductivity of solution is called **conductometric analysis**. The migration of ions is responsible for the conduction of electricity in solutions.

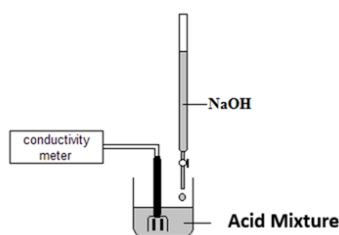
Conductance is defined as the reciprocal of resistance.

$C = 1/R$ Where **R** is resistance

The unit of conductance is ohm^{-1} or Ω^{-1} . The SI unit is Siemen(S).

Conductivity cells are usually made of quartz, which consist of two electrodes of platinum plates sealed into glass tubes. The conductivity cell is connected to conductometer.

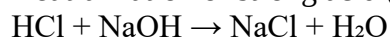
In acid–base titrations, the neutralization point is usually detected by a color change of an indicator. Alternatively, it can be determined by conductometric titration, which is based on changes in the electrical conductance of the solution during titration. Since the change in conductance near the equivalence point is not abrupt, the end point is obtained graphically by plotting conductance against the volume of titrant added. The principle of conductometric titration involves the replacement of ions having higher ionic mobility by ions of lower or different mobility during the course of the titration. In the conductometric titration of a mixture of a strong acid (HCl) and a weak acid (CH₃COOH) with a strong base (NaOH), the conductance initially decreases due to the replacement of highly mobile H⁺ ions by the less mobile Na⁺ ions. This decrease continues until all the HCl is neutralized. Subsequently, the conductance increases slowly as the weak acid (CH₃COOH) is converted into its salt, sodium acetate (CH₃COONa). On further addition of NaOH beyond the equivalence point, the conductance increases sharply due to the presence of excess Na⁺ and highly mobile OH[−] ions.



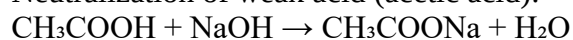
Procedure:

Fill a micro burette with the standard NaOH solution. Pipette out 50 cm³ of the given acid mixture into a clean 100 cm³ beaker. Immerse the conductivity cell in the solution and connect it to the conductometer to measure the initial conductance. Add 0.5 cm³ of NaOH to the beaker, stir gently, and record the conductance. Continue the titration by adding NaOH in successive increments of 0.5 cm³, measuring the conductance after each addition. Initially, the conductance decreases, then increases slowly, and finally increases sharply. Continue the titration until the conductance becomes nearly equal to the initial conductance value. Plot a graph of conductance (Y-axis) versus volume of NaOH added (X-axis). The point V₁ corresponds to the neutralization of HCl, while V₂ corresponds to the complete neutralization of both HCl and CH₃COOH. The difference between V₂ and V₁ gives the volume of NaOH required to neutralize CH₃COOH.

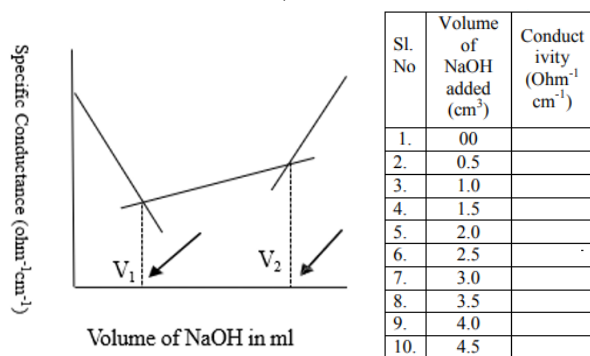
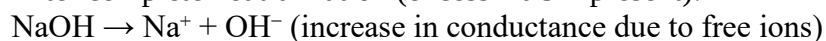
Neutralization of strong acid (HCl):



Neutralization of weak acid (acetic acid):



After complete neutralization (excess NaOH present):



Calculation

Volume of NaOH required to neutralize HCl in acid mixture

$$V_1 = \text{_____ cm}^3$$

Volume of NaOH required to neutralize CH_3COOH in acid mixture ($V_2 - V_1$) = _____ cm^3

$$\text{Normality of HCl} = \frac{(N \times V_1)_{\text{NaOH}}}{50} = \text{_____} N = (a)$$

$$\text{Weight of HCl present in the acid mixture} = N_{\text{HCl}} \times \text{Eq. wt. of HCl} = a \times 36.46 = \text{_____ g/litre}$$

$$\text{Normality of CH}_3\text{COOH} = \frac{N_{\text{NaOH}} \times (V_2 - V_1)}{50} = \text{_____} N = (b)$$

$$\text{Weight of CH}_3\text{COOH/dm}^3 = N_{\text{CH}_3\text{COOH}} \times \text{Eq. weight of CH}_3\text{COOH} = b \times 60.05 = \text{_____ g/litre}$$

Result:

$$\text{Weight of HCl present in the given acid mixture} = \text{_____ g/litre}$$

$$\text{Weight of CH}_3\text{COOH in the given acid mixture} = \text{_____ g/litre}$$

Applications of Conductometric Sensors

1. Environmental Monitoring
2. Food and Beverage Industry
3. Medical and Biomedical Applications
4. Industrial Applications
5. Agriculture
6. Energy Systems

Colorimetric Sensors

Colorimetric sensors are simple type of optical sensors which use incident light of visible region (400-800nm) of spectrum and detect and determine the concentration of colored chemical species of analyte solutions. The important parts of colorimeter are given below.

Principle: Quantitative analysis by colorimetry is based on **Beer-Lambert's law**.

When a beam of monochromatic radiation is passed through a solution of an absorbing substance. The intensity of incident radiation decreases exponentially with increase in thickness and concentration of the solution.

$$\log(I_0/I_t) = A = \epsilon Cl$$

I_0 = intensity of the incident light

I_t = intensity of transmitted light

A = absorbance

ϵ = molar absorption co-efficient

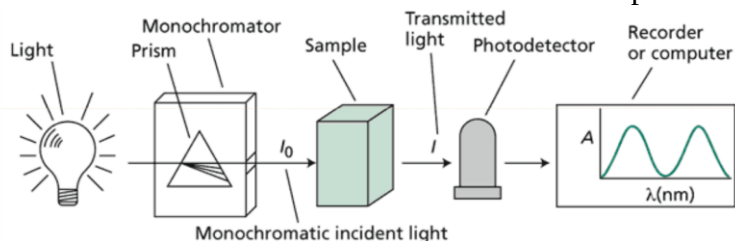
l = length of light path

C = concentration

The term, $\log(I_0/I_t)$ is called the absorbance or optical density of the light absorbing medium. Since ϵ is a constant and thickness l is kept constant.

Working:

When analyte solution interacts with an incident light of suitable wavelength, certain quantity of light is absorbed by the analyte solution, and a transducer such as photo detector converts intensity of light absorbed by analyte into electrical signal. A monochromatic light whose wavelength falls in visible region of spectrum is made to pass through analyte solution where certain quantity of incident light is absorbed and remaining amount of light is transmitted. Concentration of colored chemical species in analyte solution is determined by measuring absorbance or transmittance of light of particular wave length.



Monochromator – To select the particular wavelength filter or monochromators are used to split the light from the light source.

Sample holder – Test tube or Cuvettes are used to hold the color solutions they are made up of Glass at the visible wavelength.

Photo Detector System – when light falls on the detector system, an electric current is generated, this reflects the Galvanometer reading.

Measuring device (recorder) – The current from the detector is fed to the measuring device, the Galvanometer, shows the meter reading that is directly proportional to the intensity of light.

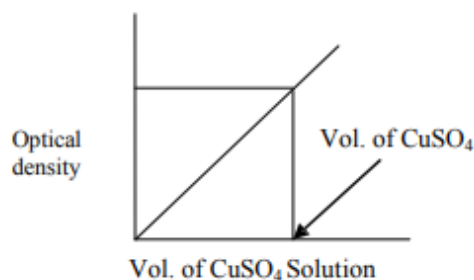
Transfer the given copper sulphate solution (stock solution) to a burette and draw out 5, 10, 15, 20 and 25 cm³ of the solution into 50 cm³ volumetric flasks. Add 5 cm³ of ammonia solution to each of them and dilute up to the mark with ion exchange water. Stopper the flasks and mix. To the test solution given in a 50 cm³ volumetric flask, add 5 cm³ of ammonia and dilute up to the mark and mix well. Prepare a blank solution by diluting 5 cm³ of ammonia solution in a 50 cm³ volumetric flask up to the mark with ion exchange water and mix well. Measure the absorbance of the solutions against blank at 620 nm using photoelectric colorimeter. Draw a calibration curve by plotting optical density (OD) against volume of copper sulphate (cm³). Using the calibration curve, find out the concentration of copper in the test solution and calculate the amount of copper in 50 cm³ of the given test solution.

A series of standard solutions containing Cu (II) ions is treated with NH₃ to get dilute solution of the complex of cuprammonium complex ion



Maximum absorbance of this complex ion is 620nm.

Flask No	Volume of NH ₃ (cm ³)	Volume of copper sulphate solution (cm ³)	Absorbance (Optical Density)
Blank (B)	5	0	
1	5	5	
2	5	10	
3	5	15	
4	5	20	
5	5	25	
TS	5	Test solution	



The reason for getting a straight line is because it follows, **Beers Lamberts Law**.

Applications of Conductometric Sensors

7. Environmental Monitoring
8. Food and Beverage Industry
9. Medical and Biomedical Applications
10. Industrial Applications
11. Agriculture
12. Energy Systems

PRINCIPLE AND INSTRUMENTATION OF POTENTIOMETRIC SENSOR

Potentiometric titrations

Potentiometric sensors: Estimation of iron

Principle: The potential of an electrode depends on the concentration of the solution. Potentiometric titrations involve in measurement of electrode potentials with addition of titrant. The electrode potential is given by Nernst equation

$$E = E^0 + (2.303RT/nF) \log[M^{n+}]$$

The potentiometric titration requires the selection of the appropriate indicator electrode and reference electrode is usually a **platinum** and **saturated calomel electrode**, respectively.

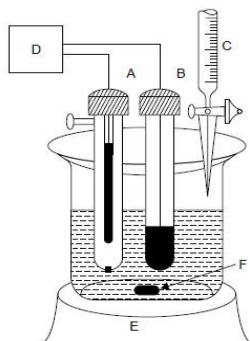
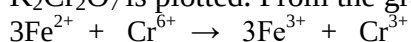


Fig. Potentiometric titration assembly

Take 25 cm³ of the given FAS solution into a clean beaker and add 2 test tubes of dil. H₂SO₄. Immerse platinum electrode and saturated calomel electrode assembly, connect to a potentiometer and note down the potential value in milli volts (mV). Fill a micro burette with the given standard K₂Cr₂O₇ solution. Add K₂Cr₂O₇ solution in increments of 0.5 cm³ to FAS solution in the beaker. Stir the solution thoroughly and measure the emf after each addition. When the reaction reaches the equivalence point, there is a maximum increase in the emf value. After the equivalence point, there will be slight increase in emf on continued addition of K₂Cr₂O₇. Graph of E/ V against volume of K₂Cr₂O₇ is plotted. From the graph, volume of K₂Cr₂O₇ at equivalence point is found out.

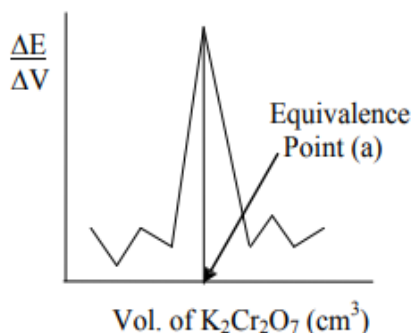


At the equivalence point, there will be sudden jump So the increase in the value of Fe³⁺/Fe²⁺ and change in the electrode from Pt/ Fe³⁺, Fe²⁺ (0.77V) to Pt/Cr⁶⁺, Cr³⁺ (1.33V), there is a huge change in potential at the end point.

$$E = E^0 + 0.0591 \log [\text{Fe}^{2+}]/[\text{Fe}^{3+}]$$

The $\Delta E/\Delta V$ plotted against the volume of K₂Cr₂O₇ added, the end point where the slope of curve is maximum.

Volume of K ₂ Cr ₂ O ₇ (cm ³)	E (mV)	ΔE	ΔV	$\frac{\Delta E}{\Delta V}$
0		NIL	NIL	NIL
0.5				
1.0				
1.5				
2.0				
2.5				
3.0				
3.5				
4.0				
4.5				
5.0				



Calculation

Normality of K₂Cr₂O₇ = N (will be given) ----- (b)

Normality of Iron = Normality of K₂Cr₂O₇ × Volume of K₂Cr₂O₇ (from graph)/Volume of Iron

$$N = b \times a / 25 = \text{-----} N (z)$$

Weight of Iron /dm³ = Normality of Iron × Equivalent weight of Iron (55.85)

$$= z \times 392 \text{ g} = \text{-----} \text{g/litre (w)}$$

Result: Weight of Iron in given solution = -----g/litre

Applications of potentiometric Sensors

1. Environmental Monitoring
2. Clinical and Biomedical Applications
3. Food and Agriculture
4. Industrial Applications
5. Pharmaceutical Applications

E-waste

Introduction

E-Waste refers to unwanted or unused material generated from the waste of Electronic and Electrical Equipment (WEEE). Alternately, all items of electrical and electronic equipment and its parts that have been discarded by the user as waste without the purpose of re-use or re-cycle is called Electronic Waste.

Definition:

Electronic waste, also called e-waste is various forms of electric and electronic equipment that have no longer satisfy their original purpose.

Types of E-Waste:

E-Waste has been categorized into the following types.

Type 1- Major appliances (refrigerators, washing machines, dryers etc.)

Type 2 – Small appliances (vacuum cleaners, irons, blenders, fryers etc.)

Type 3 – Computer and telecommunication appliances (laptops, telephones, mobile etc.)

Type 4 – Consumer electronics (video and audio equipment, musical instruments)

Type 5 – Lighting devices (incandescent light bulbs, fluorescent tubes, gas-discharge lamps)

Type 6 – Electronic tools (drills, saws, gardening devices etc.)

Type 7 – Electronic toys, sports equipment

Type 8 – Medical devices (all medical equipment)

Type 9 – Monitoring devices (detectors, thermostats, laboratory equipment etc.)

Type10 –Vending machines

Effects of E-Waste on Environment and Human Health

- **Soil contamination:** Toxic metals (lead, cadmium, mercury) seep into soil, reducing fertility and entering the food chain.
- **Water pollution:** Leaching of heavy metals and acids contaminates groundwater and surface water sources.
- **Air pollution:** Open burning of plastics and circuit boards releases dioxins, furans, and other harmful gases.
- **Loss of biodiversity:** Toxic chemicals harm aquatic life, plants, and animals, disrupting ecosystems.
- **Resource depletion:** Failure to recycle valuable metals increases dependence on mining.

On Human Health

- **Neurological damage:** Lead and mercury exposure affects brain function, especially in children.
- **Respiratory issues:** Inhalation of toxic fumes from burning e-waste causes lung diseases and breathing problems.
- **Kidney and liver damage:** Cadmium and other metals accumulate in vital organs, impairing their function.
- **Cancer risk:** Long-term exposure to carcinogenic substances like dioxins, arsenic, and brominated flame retardants.
- **Reproductive and developmental disorders:** Toxic chemicals affect fertility, pregnancy, and child development.
- **Skin and eye irritation:** Direct contact during informal recycling causes rashes, burns, and eye problems.

Methods of disposal

Non-biodegradable and toxic wastes like radioactive remnants can potentially cause irreparable damage to the environment and human health if not strategically disposed of. Waste disposal has been a matter of concern, the main problem growth in population and industrialization. Here are a few methods of waste disposal.

Disposal of E-waste are mainly done in 5 types.

1. Recycling
2. Reuse
3. Metal recovery by acid
4. Insinuation
5. Secured land-filling

1. Recycling: Devices such as electronic waste, monitors, picture tubes, laptops, keyboards, telephones, hard drives, CD drives, fax machines, printers, CPUs, modem cables etc. can be recycled. In this process, various metals and plastics are sabotaged separately and preserved for reuse.

2. Reuse: The metals from the old electronic devices can be extracted and reused. Such as from computers, mobiles, laptops etc.

3. Metal recovery by acid: Different types of parts like ferrous and non ferrous metal and printed circuit board are separated by electronic waste. Different types of metals like lead, copper, aluminum, silver, gold, platinum etc. are recovered and reused.

4. Incineration: In this process the e-waste is lit in a fully closed chamber inside the insulator at a temperature of 900 to 1000 degree centigrade. Due to which the quantity of e-waste is reduced considerably and the toxicity of the organic substance present in it is reduced significantly. The smoke and gas coming out of the chimney in the injector is passed through the Air Pollution Control System (APCS) and the various types of metals present in the smoke are separated by chemical action and the gases are treated.

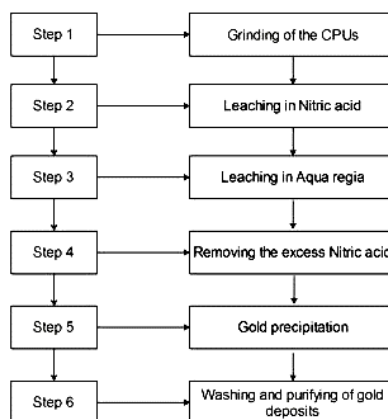
5. Secured land-filling: The e-waste is built on the flat land and the pits are pressed into the soil by putting the e-waste in it. Residues from waste incineration or waste that is not suitable for material recycling or thermal treatment are deposited in landfills.

Artificial Intelligence (AI) in E-Waste Management

- **Identification of hazardous chemicals:** AI tools detect and classify toxic chemicals such as lead, mercury, cadmium, and brominated flame retardants in e-waste using advanced sensor data and pattern recognition.
- **Chemical composition analysis:** Machine learning models process spectral data (XRD, XRF, FTIR) to rapidly determine the composition of metals, plastics, and semiconductors, enabling targeted recycling.
- **Intelligent sorting and separation:** AI-driven robotic systems utilize chemical property data (conductivity, density, magnetic behavior) to efficiently separate recyclable metals from hazardous or non-recyclable substances.
- **Prediction of toxic leaching:** AI predicts the leaching behaviour of chemicals from e waste into soil and water, supporting the design of eco-friendly chemical treatments and containment strategies.
- **Optimization of recycling processes:** Algorithms recommend green chemical methods such as bioleaching and solvent extraction for efficient recovery of valuable metals like gold, copper, and palladium.
- **Environmental impact monitoring:** AI models continuously track chemical pollution levels in air, water, and soil around e-waste sites, ensuring compliance with environmental safety standards.
- **Waste-to-resource conversion:** AI supports chemical upcycling, transforming hazardous e-waste materials into valuable products such as nanomaterials and catalysts.
- **Data-driven policy and management:** AI systems provide actionable insights for policymakers to regulate toxic chemical disposal and promote sustainable, advanced recycling technologies.

Extraction of Gold from e-waste

Recovering precious metals from e-waste through hydrometallurgical processes is more attractive economically than other methods. A hydrometallurgical process consists of first a set of operations including acidic or alkaline dissolution (leaching) of solids. The resulting solutions are then exposed to separation and purification methods including deposition, solvent extraction, adsorption, and ion exchange in order to isolate and concentrate the intended metals.



- **Step 1:** (grinding of the CPUs): First, any dust or other particles were removed from the CPUs. The size of the grind pieces was 1 mm.
- **Step 2:** (leaching in nitric acid): Concentrated nitric acid (65%) was added. Leaching time in this stage was 1 hour.
- **Step 3:** (leaching in aqua regia): The aqua regia solution was obtained using three volumes of HCl plus one volume of HNO₃ (under the hood). Typically, the gold of composite CPU-containing boards is easily dissolved in 4-to-1 aqua regia. The solution was decanted, and the pieces were subsequently washed with 10 mL of concentrated HCl so that all of the gold content was fed into the solution.
- **Step 4:** (removing the excess nitric acid): Sulfuric acid was added to the solution to accelerate the removal of nitric acid and sedimentation of the lead as lead sulfate (if present). The time required for the complete removal of nitric acid from this solution was 45 minutes.
- **Step 5:** (precipitation of gold): A certain amount of iron sulfate (per one gram of gold: 4.2 g of iron sulfate) was dissolved in warm water and gradually added to the gold-containing solution. For better dissolution of iron sulfate, some drops of HCl can be used.
- **Step 6:** (washing and purifying gold deposits): The deposit was coated with HCl and then boiled (to dissolve the excess copper sulfate). Following boiling, the acid was removed and the deposit was re-washed in HCl to ensure the cleanness of the gold powder. Eventually, the deposit was filtered and washed with large amounts of warm water. At this stage, the deposit was placed in an oven at a temperature of 100°C to dry. Sampling and measuring of precious metals were performed following the first and second stages of leaching, after the stage of gold extraction, and finally following the purification of deposits.

Questions

1. Explain Synthesis, properties and uses in touch screen of Polylactic Acid (PLA).
2. Describe Synthesis, properties and uses in touch screen of Polyethylene Glycol (PEG).
3. Write the Synthesis and properties of Alginate Hydrogel and its uses in Brain Computer Interfaces (BCIs) applications.
4. Explain the Principle, Construction and Working of Glucose Biosensor.
5. Explain the Principle, Construction, and Application in Acid Mixture using Conductometric Sensors.
6. Explain the principle, instrumentation and its application in the estimation of copper Colorimetric sensor.
7. Explain the principle and instrumentation of potentiometric sensor.
8. Write the types of E-Waste.
9. Explain effects of e-waste on environment and human health.
10. Describe the methods of disposal of E-waste.
11. Write the utility of Artificial Intelligence (AI) in E-Waste Management.
12. Explain the steps involved in the extraction of gold from e-waste.

*****END*****